

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, SQL, C++, Bash

LLM & Agentic Systems

LLM agents, end-to-end agent pipelines, tool use, provider routing/fallback, prompt design, evaluation, OpenAI / Groq APIs, Ollama, MCP

Retrieval & RAG

RAG, RRF, cross-encoder reranking, MMR, knowledge graphs, knapsack token budgeting, Brave Search, web scraping, cell-level provenance, sentence-transformers

ML & Deep Learning

PyTorch, JAX, HuggingFace Transformers, NumPy, autodiff, diffusion

Backend & Cloud

FastAPI, Node.js, Express, REST APIs, GraphQL, async services, Docker, AWS, GCP, Azure, Cloudflare Workers, Git, GitHub Actions

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies from conversation history and user input; iterated on prompt design and response-quality heuristics for a real customer-facing experience.
- Designed and shipped **20+ RESTful APIs** in Node.js + Fastify over PostgreSQL with input validation and clear contracts, and migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and maintainability.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **agentic** web-discovery service that converts open-ended topic queries into structured entity tables via typed retrieval planning, page rerank, deterministic extraction with LLM fallback, output verification, and **cell-level provenance** for every result; deployed behind a FastAPI service with a live hosted demo.
- Hardened the agent with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

KG2RAG-Enhanced - Knowledge-Graph-Guided RAG for Multi-Hop QA

github.com/Deva-1903/KG2RAG-685-NLP

Python, LLaMA-3 (Ollama), sentence-transformers, networkx, spaCy

- Built the multi-view retrieval component of a team **KG-guided RAG** pipeline for HotpotQA, decomposing complex questions into single-hop sub-questions and fusing per-view retrieval with **RRF**, cross-encoder reranking, and **MMR**, lifting supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG2RAG baseline on a **4,905-question** evaluation.
- Grounded retrieval in a team-built **knowledge graph** spanning **28,492 entities** with token-budgeted evidence selection; ran 4,905-question and controlled N=1,000 batch experiments with token and latency logging.

cf-ai-research-scout - Edge-Native AI Research Agent

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built and deployed an edge-native AI research **agent end to end** on Cloudflare Workers: it fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time.
- Designed durable workflow boundaries (WorkflowEntrypoint + Durable Objects) so multi-step agent tasks survive process restarts and tool failures, persisting agent state in D1 without an external database.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.